

Computational Sociology

Online experiments and surveys

Dr. Thomas Davidson

University of North Carolina at Chapel Hill

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Course updates

Homework

- ▶ Homework 1 due next Friday, September 18
 - ▶ Two parts
 - ▶ Data structures and programming
 - ▶ APIs and web scraping
 - ▶ Submit by committing and pushing to your GitHub homework repository
 - ▶ Unlimited commits before the deadline
 - ▶ Further instructions in the GitHub setup wiki on the course website

Plan

- ▶ Part I: Online experiments
- ▶ Part II: Online surveys

Part I: Online experiments

Online experiments

Motivation

- ▶ Lab experiments offer control over the setting and treatment
 - ▶ Findings may not generalize to other populations or settings
- ▶ Field experiments test interventions in everyday settings
 - ▶ Random assignment supports causal inference, though exposure and spillovers can be harder to control
- ▶ Digital experiments can combine random assignment, realistic settings, and large samples

Online experiments

Methods: Internal experiments

- ▶ Companies and other actors experiment internally
 - ▶ A/B tests compare alternative interfaces and product features
- ▶ The vast majority of these experiments are private, but some are published by researchers
- ▶ Thousands of experiments are available in the Upworthy Research Archive: <https://upworthy.natematias.com/>

Online experiments

Emotional contagion on Facebook

PNAS PNAS PNAS PNAS PNAS

Experimental evidence of massive-scale emotional contagion through social networks

Adam D. I. Kramer^{a,1}, Jamie E. Guillory^{b,2}, and Jeffrey T. Hancock^{b,c}

^aCore Data Science Team, Facebook, Inc., Menlo Park, CA 94025; and Departments of ^bCommunication and ^cInformation Science, Cornell University, Ithaca, NY 14853

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved March 25, 2014 (received for review October 23, 2013)

Emotional states can be transferred to others via emotional contagion, leading people to experience the same emotions without their awareness. Emotional contagion is well established in laboratory experiments, with people transferring positive and negative emotions to others. Data from a large real-world social network, collected over a 20-y period suggests that longer-lasting moods (e.g., depression, happiness) can be transferred through networks [Fowler JH, Christakis NA (2008) *BMJ* 337:a2338], although the results are controversial. In an experiment with people who use Facebook, we test whether emotional contagion occurs outside of in-person interaction between individuals by reducing the amount of emotional content in the News Feed. When positive expressions were reduced, people produced fewer positive posts and more negative posts; when negative expressions were reduced, the opposite pattern occurred. These results indicate that emotions expressed by others on Facebook influence our own emotions, constituting experimental evidence for massive-scale contagion via social networks. This work also suggests that, in contrast to prevailing assumptions, in-person interaction and non-verbal cues are not strictly necessary for emotional contagion, and that the observation of others' positive experiences constitutes a positive experience for people.

computer-mediated communication | social media | big data



demonstrated that (i) emotional contagion occurs via text-based computer-mediated communication (7); (ii) contagion of psychological and physiological qualities has been suggested based on correlational data for social networks generally (7, 8); and (iii) people's emotional expressions on Facebook predict friends' emotional expressions, even days later (7) (although some shared experiences may in fact last several days). To date, however, there is no experimental evidence that emotions or moods are contagious in the absence of direct interaction between experimenter and target.

On Facebook, people frequently express emotions, which are later seen by their friends via Facebook's "News Feed" product (8). Because people's friends frequently produce much more content than one person can view, the News Feed filters posts, stories, and activities undertaken by friends. News Feed is the primary manner by which people see content that friends share. Which content is shown or omitted in the News Feed is determined via a ranking algorithm that Facebook continually develops and tests in the interest of showing viewers the content they will find most relevant and engaging. One such test is reported in this study: A test of whether posts with emotional content are more engaging.

The experiment manipulated the extent to which people ($N = 689,003$) were exposed to emotional expressions in their News Feed. This tested whether exposure to emotions led people to

Online experiments

Emotional tone of feed affected users' language

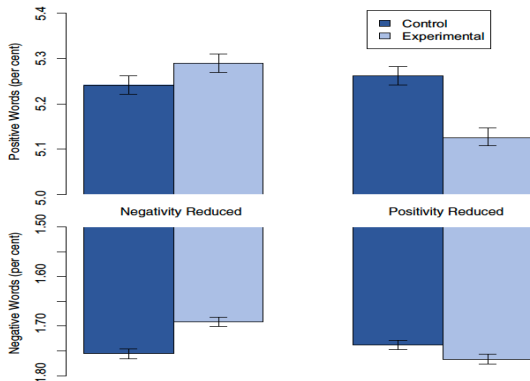


Fig. 1. Mean number of positive (*Upper*) and negative (*Lower*) emotion words (percent) generated people, by condition. Bars represent standard errors.

Online experiments

Reactions



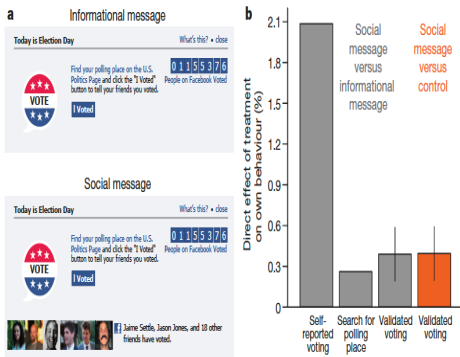
<https://www.cbsnews.com/news/controversial-facebook-emotion-study-journal-responds/>

Reactions

Computational Sociology

Online experiments

Social influence and voter turnout



Bond, Robert M., Christopher J. Fariss, Jason J. Jones, Adam D. I. Kramer, Cameron Marlow, Jaime E. Settle, and James H. Fowler. 2012. "A 61-Million-Person Experiment in Social Influence and Political Mobilization." *Nature* 489 (7415): 295-98.

Online experiments

Platform research during the 2020 election

SOCIAL MEDIA

How do social media feed algorithms affect attitudes and behavior in an election campaign?

Andrew M. Guess^{1*}, Neil Malhotra², Jennifer Pan³, Pablo Barberá⁴, Hunt Allcott⁵, Taylor Brown⁴, Adriana Crespo-Tenorio⁴, Drew Dimmery^{4,6}, Deen Freelon⁷, Matthew Gentzkow⁸, Sandra González-Bailón⁹, Edward Kennedy¹⁰, Young Mie Kim¹¹, David Lazer¹², Devra Moehler⁴, Brendan Nyhan¹³, Carlos Velasco Rivera⁴, Jaime Settle¹⁴, Daniel Robert Thomas⁴, Emily Thorson¹⁵, Rebekah Tromble¹⁶, Arjun Wilkins⁴, Magdalena Wojcieszak^{17,18}, Beixian Xiong⁴, Chad Kiewiet de Jonge⁴, Annie Franco⁴, Winter Mason⁴, Natalie Jomini Stroud¹⁹, Joshua A. Tucker²⁰

We investigated the effects of Facebook's and Instagram's feed algorithms during the 2020 US election. We assigned a sample of consenting users to reverse-chronologically-ordered feeds instead of the default algorithms. Moving users out of algorithmic feeds substantially decreased the time they spent on the platforms and their activity. The chronological feed also affected exposure to content: The amount of political and untrustworthy content they saw increased on both platforms, the amount of content classified as uncivil or containing slur words they saw decreased on Facebook, and the amount of content from moderate friends and sources with ideologically mixed audiences they saw increased on Facebook. Despite these substantial changes in users' on-platform experience, the chronological feed did not significantly alter levels of issue polarization, affective polarization, political knowledge, or other key attitudes during the 3-month study period.

Guess, Andrew M. et al. 2023. "How Do Social Media Feed Algorithms Affect Attitudes and Behavior in an Election Campaign?" *Science* 381(6656):398-404.

Online experiments

Methods: Using existing environments

- ▶ Researchers can use platforms to create their own experiments
 - ▶ e.g. Doleac and Stein (2013) used different pictures on Craigslist to measure discrimination
 - ▶ e.g. van de Rijt et al. (2014) randomly donated to Kickstarters, upvoted reviews, awarded Wikipedia contributors, and signed petitions to study the Matthew Effect
 - ▶ e.g. Munger (2017) used a Twitter “bot” to measure the effect of sanctions on racial harassment
 - ▶ e.g. Bail et al. (2018) paid Twitter users to follow a bot that retweeted the other side

Countering hate speech on Twitter

Polit Behav
DOI 10.1007/s11109-016-9373-5



ORIGINAL PAPER

Tweetment Effects on the Tweeted: Experimentally Reducing Racist Harassment

Kevin Munger¹

Online experiments

Design and experimental manipulation



Fig. 3 Treatments. **a** The treatment—black bot. **b** The bot applying the treatment—white bot

Online experiments

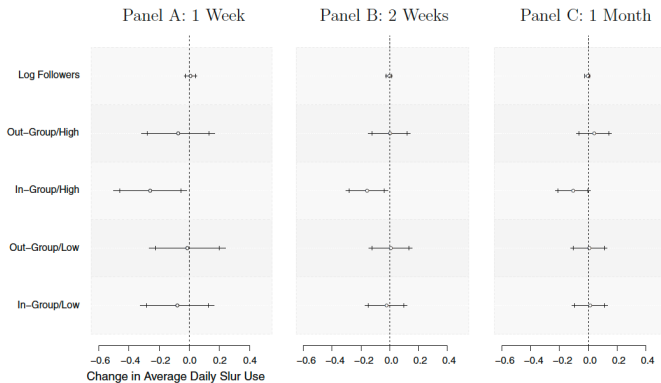
Hypotheses

Table 1 Experimental design and hypothesized effect sizes

	In-group	Out-group
Low followers	Medium effect	Small effect
High followers	Large effect	Medium effect

Online experiments

Results



Echo chambers and exposure to opposing views

Exposure to opposing views on social media can increase political polarization

Christopher A. Bail^{a,1}, Lisa P. Argyle^b, Taylor W. Brown^a, John P. Bumpus^a, Haoan Chen^c, M. B. Fallin Hunzaker^d, Jaemin Lee^e, Marcus Mann^a, Friedolin Merhout^a, and Alexander Volfovsky^a

^aDepartment of Sociology, Duke University, Durham, NC 27708; ^bDepartment of Political Science, Brigham Young University, Provo, UT 84602; ^cDepartment of Political Science, Duke University, Durham, NC 27708; ^dDepartment of Sociology, New York University, New York, NY 10012; and ^eDepartment of Statistical Science, Duke University, Durham, NC 27708

Edited by Peter S. Bearman, Columbia University, New York, NY, and approved August 9, 2018 (received for review March 20, 2018)

There is mounting concern that social media sites contribute to political polarization by creating “echo chambers” that insulate people from opposing views about current events. We surveyed a large sample of Democrats and Republicans who visit Twitter at least three times each week about a range of social policy issues. One week later, we randomly assigned respondents to a treatment condition in which they were offered financial incentives to follow a Twitter bot for 1 month that exposed them to messages from those with opposing political ideologies (e.g., elected officials, opinion leaders, media organizations, and nonprofit groups). Respondents were resurveyed at the end of the month to measure the effect of this treatment, and at regular intervals throughout the study period to monitor treatment compliance. We find that Republicans who followed a liberal Twitter bot became substantially more conservative posttreatment. Democrats exhibited slight increases in liberal attitudes after following a conservative Twitter bot, although these effects are not statistically significant. Notwithstanding important limitations of our study, these findings have significant implications for the interdisciplinary literature on political polarization and the emerging field of computational social science.

challenges for the study of social media echo chambers and political polarization, since it is notoriously difficult to establish whether social media networks shape political opinions, or vice versa (27–29).

Here, we report the results of a large field experiment designed to examine whether disrupting selective exposure to partisan information among Twitter users shapes their political attitudes. Our research is governed by three preregistered hypotheses. The first hypothesis is that disrupting selective exposure to partisan information will decrease political polarization because of intergroup contact effects. A vast literature indicates contact between opposing groups can challenge stereotypes that develop in the absence of positive interactions between them (30). Studies also indicate intergroup contact increases the likelihood of deliberation and political compromise (31–33). However, all of these previous studies examine interpersonal contact between members of rival groups. In contrast, our experiment creates virtual contact between members of the public and opinion leaders from the opposing political party on a social media site. It is not yet known whether such virtual contact creates the

Significance

Social media sites are often blamed for exacerbating political polarization by creating “echo chambers” that insulate people

political polarization | computational social science | social networks | social media | sociology

Online experiments

Echo chambers and exposure to opposing views

- ▶ Online polarization attributed to partisan “echo chambers”
- ▶ Does exposure to opposing views moderate attitudes or reinforce existing beliefs?
- ▶ People choose who to follow (selection), making it difficult to disentangle influence using observational data

Online experiments

Research design

- ▶ Recruited self-identified Democrats and Republicans who used Twitter at least three times a week
 - ▶ Control group had no intervention
 - ▶ Treatment group asked to follow bots that retweeted 24 messages daily from opposing political elites and organizations
 - ▶ Exposure consists of elite messages rather than exchanges with out-party members

Online experiments

Research design

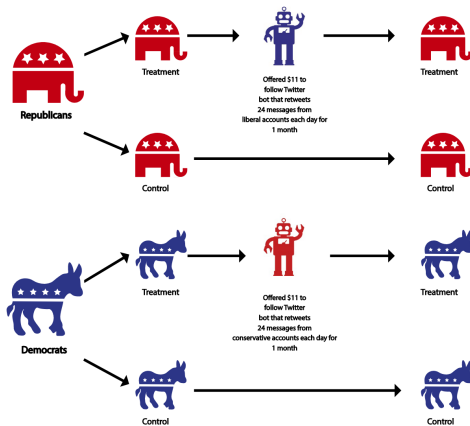


Fig. 1. Overview of research design.

Online experiments

Building the bots

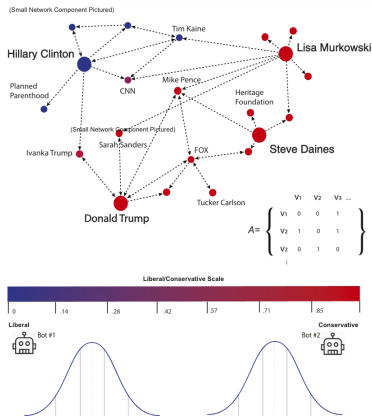
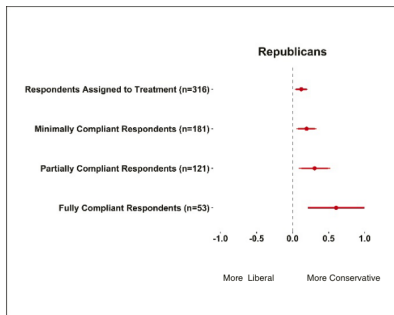
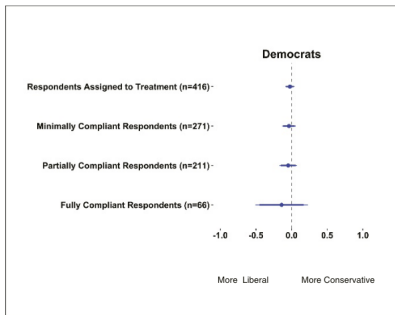


Fig. 2. Design of study's Twitter bots.

Online experiments

Asymmetric attitudinal polarization



Online experiments

Methods: Digital labs

- ▶ Create a virtual environment, fully controlled by the researcher
- ▶ High fixed costs associated with developing a platform
- ▶ Near zero variable cost experiments
 - ▶ But incentivizing participation is challenging

Online experiments

Explaining success in creative industries

- ▶ Creative fields like music and arts are often characterized by inequality and winner-take-all dynamics, yet experts routinely fail to predict which products will succeed
- ▶ Competing explanations
 - ▶ The best products really are qualitatively different from the rest but hard to predict
 - ▶ Success is a function of social influence and market behavior

Online experiments

The Music Lab

Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market

Matthew J. Salganik,^{1,2*} Peter Sheridan Dodds,^{2*} Duncan J. Watts^{1,2,3*}

Hit songs, books, and movies are many times more successful than average, suggesting that “the best” alternatives are qualitatively different from “the rest”; yet experts routinely fail to predict which products will succeed. We investigated this paradox experimentally, by creating an artificial “music market” in which 14,341 participants downloaded previously unknown songs either with or without knowledge of previous participants’ choices. Increasing the strength of social influence increased both inequality and unpredictability of success. Success was also only partly determined by quality: The best songs rarely did poorly, and the worst rarely did well, but any other result was possible.

Online experiments

Research design

- ▶ Participants ($n=14,341$) could listen to 48 songs by unknown bands
- ▶ Independent condition: Song and band names only
- ▶ Social influence condition: Number of previous downloads added
- ▶ Ranking condition (only for social influence): whether or not songs were randomized (Exp.1) or ranked by downloads (Exp.2)

Online experiments

Music Lab - Song Selection - Mozilla Firefox

File Edit View Go Bookmarks Tools Help

http://www.musiclab.columbia.edu/me/control/

	[Help]	[Log off]	# of down loads
PARKER THEORY: "the sad"			159
THE FASTLANE: "if death do us part (i dont)"			108
SELSIUS: "stars of the city"			62
STUNT MONKEY: "inside out"			56
BY NOVEMBER: "if i could take you"			55
FORTHFADING: "leap"			49
HYDRAULIC SANDWICH: "separation anxiety"			43
SILENT FILM: "all i have to say"			40
UNDO: "while the world passes"			36
BENEFIT OF A DOUBT: "run away"			32
A BLINDING SILENCE: "monsters and miracles"			27
MISS OCTOBER: "pink aggression"			26
STAR CLIMBER: "tell me"			24
FAR FROM KNOWN: "route 5"			22
HALL OF FAME: "best mistakes"			21
EMBER SKY: "this upcoming winter"			19

Done

Online experiments

Parallel worlds

- ▶ Participants randomly assigned to either independent condition or one of eight worlds
 - ▶ Each world started with the same songs and zero downloads, and popularity rankings developed independently

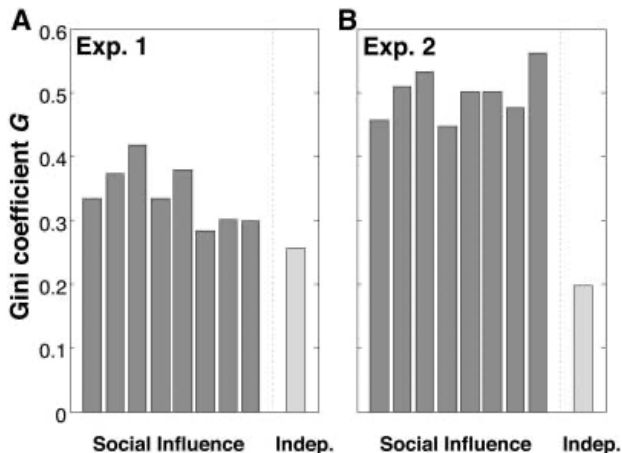
Online experiments

Measuring outcomes

- ▶ Quality: Market share in the independent condition
- ▶ Inequality: Unequal distribution of downloads in a world (Gini coefficient)
- ▶ Unpredictability: Difference in market share between worlds

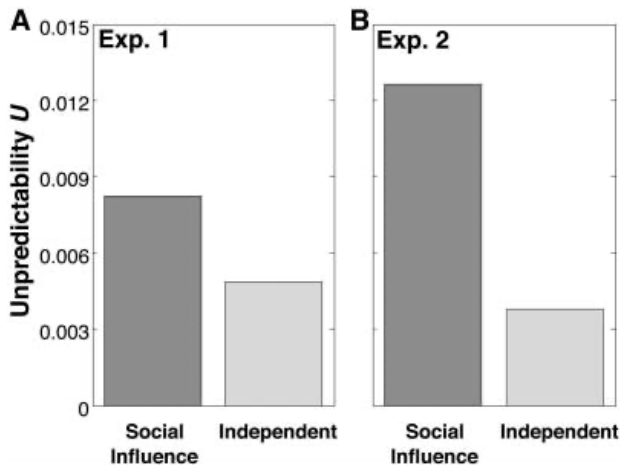
Online experiments

Social influence increases inequality



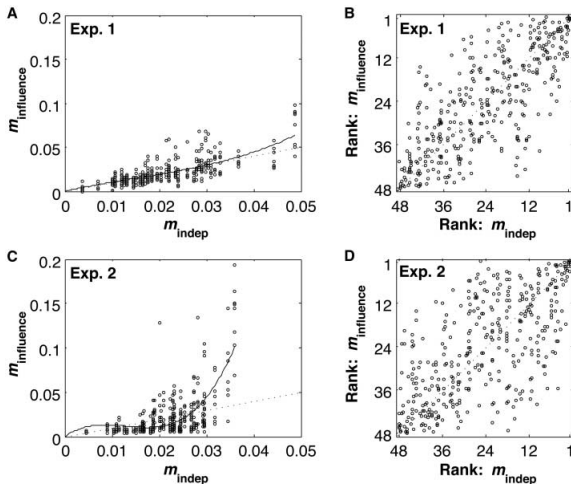
Online experiments

Social influence increases unpredictability



Online experiments

Quality still matters and ranking magnifies influence



Online experiments

Leading the herd astray

- ▶ Salganik and Watts (2008) ran another experiment where they *inverted* the download counts shown to participants so the least popular songs were presented as the most popular
 - ▶ For most songs the false popularity became a self-fulfilling prophecy
 - ▶ But the best songs recovered over time

Salganik, Matthew J., and Duncan J. Watts. 2008. "Leading the Herd Astray: An Experimental Study of Self-Fulfilling Prophecies in an Artificial Cultural Market." *Social Psychology Quarterly* 71(4):338-55.

Online experiments

Popularity before and after the inversion

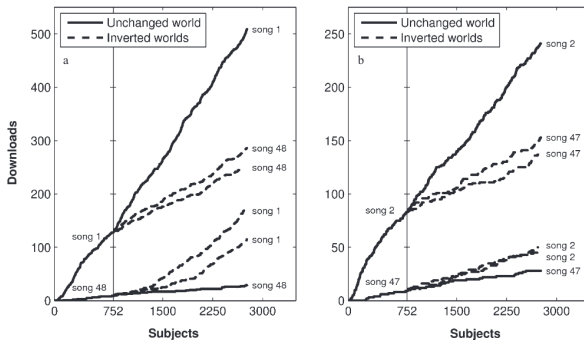
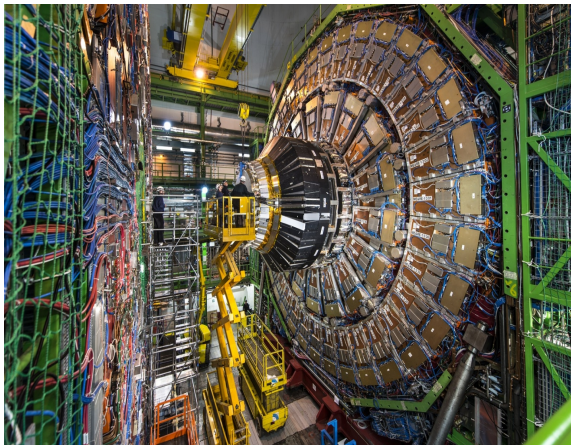


Figure 5. Popularity Dynamics Before and After the Inversion for the Most and Least Popular Song

Online experiments

New directions in digital labs



Bail, Christopher et al., 2023. "Do We Need a Social Media Accelerator?" *SocArxiv*.

Online experiments

Ethics

- ▶ Digital experimentation forces us to pay more attention to ethics
- ▶ Salganik proposes the “three R’s”
 - ▶ *Replace* experiments with less invasive methods, where possible
 - ▶ *Refine* treatment to reduce potential harm
 - ▶ *Reduce* number of participants; use the smallest sample size necessary

Part II: Online surveys

Online surveys

Three eras of survey sampling

- ▶ Area probability sampling
 - ▶ Face-to-face interviews
- ▶ Random digit dialling
 - ▶ Phone interviews
- ▶ Non-probability sampling
 - ▶ Online opt-in surveys
 - ▶ Linked “big data”

Online surveys

Issues with opt-in online sampling

- ▶ No sampling frame
- ▶ Non-representative populations
- ▶ Selection bias
- ▶ Violations of the IID assumption (e.g. snowball sampling)

Online surveys

Forecasting elections with non-representative polls



Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast



Forecasting elections with non-representative polls



Wei Wang^{a,*}, David Rothschild^b, Sharad Goel^b, Andrew Gelman^{a,c}

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ARTICLE INFO

Keywords:

Non-representative polling
Multilevel regression and poststratification
Election forecasting

ABSTRACT

Election forecasts have traditionally been based on representative polls, in which randomly sampled individuals are asked who they intend to vote for. While representative polling has historically proven to be quite effective, it comes at considerable costs of time and money. Moreover, as response rates have declined over the past several decades, the statistical benefits of representative sampling have diminished. In this paper, we show that, with proper statistical adjustment, non-representative polls can be used to generate accurate election forecasts, and that this can often be achieved faster and at a lesser expense than traditional survey methods. We demonstrate this approach by creating forecasts from a novel and highly non-representative survey dataset: a series of daily voter intention polls for the 2012 presidential election conducted on the Xbox gaming platform. After adjusting the Xbox responses via multilevel regression and poststratification, we obtain estimates which are in line with the forecasts from leading poll analysts, which were based on aggregating hundreds of traditional polls conducted during the election cycle. We conclude by arguing that non-representative polling shows promise not only for election forecasting, but also for measuring public opinion on a broad range of social, economic and cultural issues.

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Online surveys

Survey design

If the election were held today, who would you vote for?

☒ Barack Obama

☐ Mitt Romney

☐ Other

☐ Not sure

Take this one-time survey and then tell us what you think.

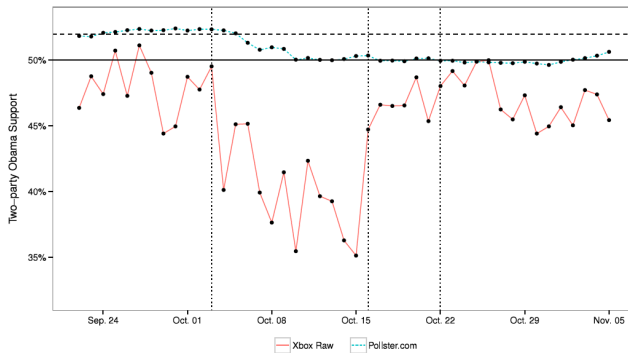
Don't worry - We'll keep your answers private and never share them with anyone else. Take a new poll each day. Thanks for giving us your view.

☒ **Get Started**

Seniors 2012 on Race (US polling) is a partnership between News and our polling partner YouGov. We will collect and store your poll answers and responses to demographic questions anonymously; we won't know how you voted or associate the data with you or your identifying information.

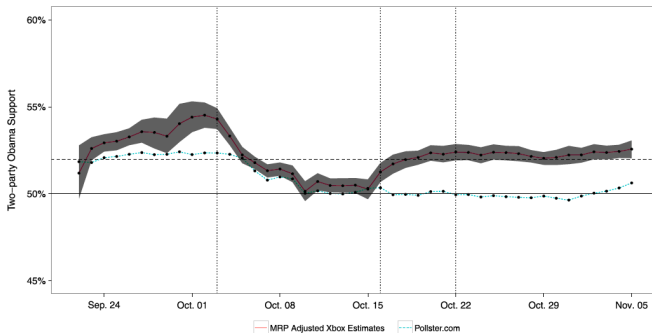
Online surveys

Raw poll data



Online surveys

Polls after adjustment



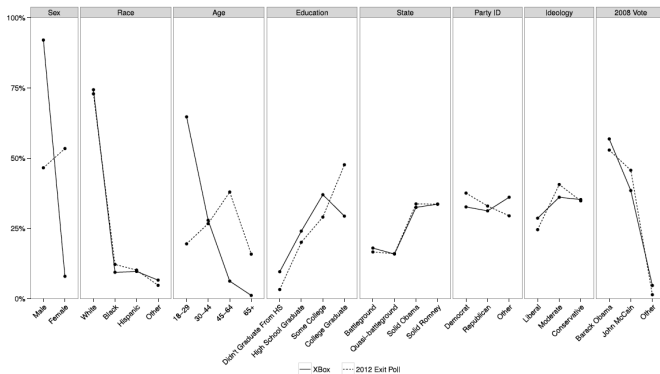
Multilevel regression and post-stratification. See Salganik 130-6 for mathematical intuition.

Monica Alexander's MRP primer with R code: <https://www.monicaalexander.com/posts/2019-08-07-mrp/>

Rohan Alexander's chapter in *Telling Stories with Data* (2023): <https://tellingstorieswithdata.com/15-mrp.html>

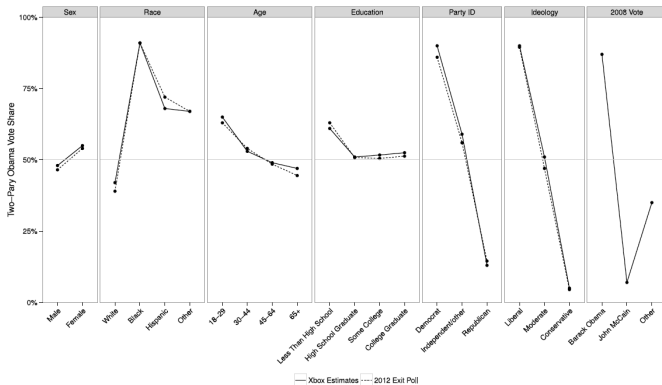
Online surveys

Demographics of Xbox users versus voters



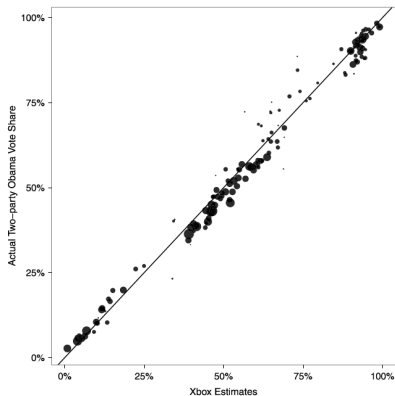
Online surveys

Population sub-group estimates



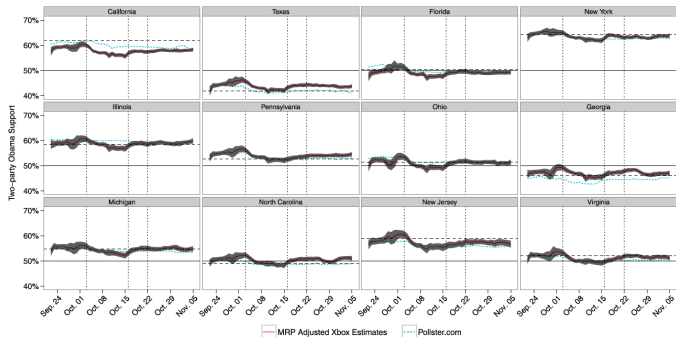
Online surveys

Prediction errors



Online surveys

State-level estimates



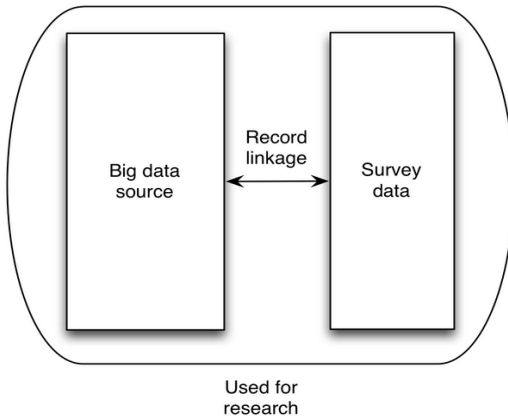
Online surveys

Working with non-probability samples

- ▶ Cheaper than fielding nationally-representative polls
- ▶ But more difficult to work with than conventional survey data
 - ▶ New statistical procedures and data sources make non-probability sampling viable

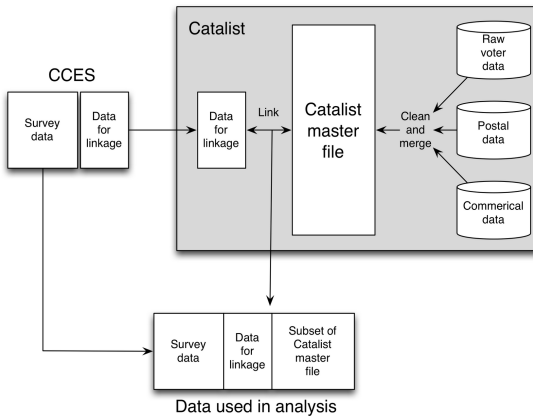
Online surveys

Record linkage / “enriched asking”



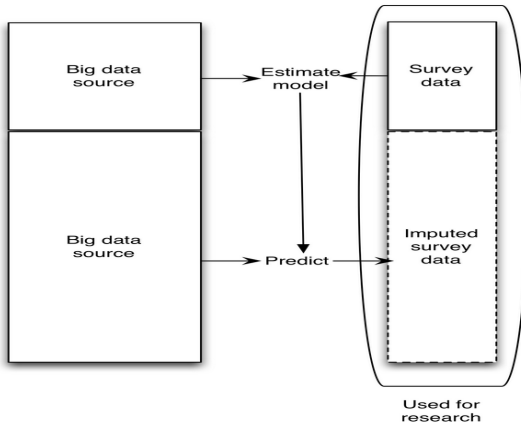
Online surveys

Enriched asking: voter behavior



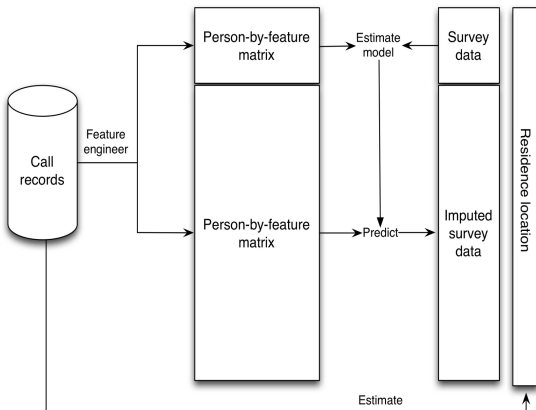
Online surveys

Big data imputation / “amplified asking”



Online surveys

Amplified asking: Mapping poverty in Rwanda



Concluding remarks

- ▶ New technologies and data sources allow us to reinvent existing methods
 - ▶ Innovative work combines social scientific approaches, statistics, and programming in new ways
- ▶ Digital experiments and surveys open up many opportunities for social scientific research
 - ▶ These methods come with challenges and require different skills from those used in conventional methods
 - ▶ New questions about ethics, informed consent, beneficence, and power

Next week

- ▶ Fundamentals of natural language processing